

The Cost of Unstructured Inventory Data in Multi-Branch Retail

An evidence review of inventory record inaccuracy, and what a structured relational system changes

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HOW TO READ THE NUMBERS IN THIS NOTE

Every figure in this report is a **published industry benchmark from a named third-party source**. None of them is a measured outcome of the AC51049 database project. The project was a design and implementation exercise for a five-branch retailer; it was not deployed into a live estate, and no sales or accuracy uplift was measured. Section 6 builds an illustrative model from published benchmarks and is labelled as such throughout.

1. EXECUTIVE SUMMARY

Retail inventory records are far less accurate than most operators assume, and the gap is expensive. The largest European field study on the subject audited roughly one million stock-keeping units across 100 stores and seven retailers and found that **59.5% of audited SKUs carried a discrepancy between the physical stock on the shelf and the number in the system**.¹ When those records were corrected, sales of the affected items rose by **5.98% on average**, and by more than 14% for the worst-affected products.¹

At a market level, IHL Group put the global cost of inventory distortion, meaning stockouts and overstocks together, at **\$1.73 trillion in 2025, equal to about 6.5% of global retail sales**.³ Gartner separately estimates that poor data quality costs the average organisation **\$12.9 million per year**.⁵

The obvious question is why the problem persists. The answer is that most operators are not running a system that can prevent it. A March 2026 survey of 400 inventory and operations professionals found that **74.2% use spreadsheets as their sole or primary inventory tool with no dedicated software behind them**, and 21.8% combine spreadsheets with paper records.⁴ In that same population, 44% experience stockouts at least monthly.⁴

This note sets out the evidence, then explains which specific failure modes a normalised relational database removes and which it does not. The central argument is narrow and defensible: **a relational model with enforced referential integrity does not by itself make inventory accurate, but it removes the class of errors that makes accuracy impossible to maintain across multiple branches**. That distinction matters, and Section 7 is explicit about the limits.

2. HOW INACCURATE ARE RETAIL INVENTORY RECORDS

2.1 The ECR field study

The strongest available evidence comes from a three-year programme run under ECR Retail Loss by Rekik, Syntetos and Glock, published in 2020.¹ The design is unusually rigorous for industry research: matched test and control store pairs, a 12-week pre-stocktake observation window, a stocktake performed in test stores only, and a 12-week post-

stocktake tracking period. Roughly one million SKUs were examined, with 233,000 used in the final analysis, across seven European retailers and 100 stores in grocery, general merchandise and fashion.

FINDING	VALUE
SKUs with a discrepancy between physical and system stock	59.54%
Grocery and general merchandise	63.39%
Fashion and apparel	54.08%
Range across individual retailers	6.20% to 73.33%
Average sales lift after records were corrected	5.98%
Range across retailers	3.83% to 8.38%
High-discrepancy products	14%+
Low-discrepancy products	2.11%

Two details in that table deserve emphasis. First, the 6.20% to 73.33% range across retailers is almost as informative as the average: this is not a uniform industry condition, it is an operational capability that some retailers have and others do not. Second, the uplift concentrates in fast-moving A-class items, which is where the commercial damage of a wrong number is greatest.

2.2 Peer-reviewed corroboration

Shabani, Maroti, de Leeuw and Dullaert examined inventory record inaccuracy at 81 stores of an international fashion retailer, covering more than 5.25 million inventory items, using a network data envelopment analysis model.² Their central result is a useful corrective to a naive reading of the ECR numbers: the performance gain from improving record accuracy is **small for stores that are already operating near efficiency and large for stores that are highly inefficient**. Accuracy work is not uniformly valuable. It pays where the operation is weak.

2.3 Accuracy decays

The ECR programme also ran a second experiment tracking 27,000 SKUs through sequential stocktakes over 16 weeks, specifically to measure how quickly accuracy deteriorates after a count.¹ This is the finding most often ignored in vendor material. A stocktake is a point-in-time correction, not a fix. Without a system that maintains record integrity between counts, the discrepancy rate climbs back. That is the argument for structural change rather than more frequent counting.

3. WHAT INVENTORY DISTORTION COSTS

IHL Group's 2025 study estimates global inventory distortion at **\$1.73 trillion**, roughly 6.5% of global retail sales.³

BREAKDOWN	2025 ESTIMATE
Global total	\$1.73tn
Asia-Pacific	\$642bn (37%)
North America	\$415bn
Attributed to supply chain disruption	\$301bn
Attributed to personnel issues	\$248bn
Attributed to organised retail crime and theft	\$379bn
EMEA improvement since 2020	31.1%

IHL is an analyst firm with vendor relationships, so the headline number should be treated as an order-of-magnitude indicator rather than a precise measurement. What survives that caveat is the composition: the largest single attributed cause is not theft or supply chain shock but the combination of process and personnel factors, which is to say, the way the operation records and acts on its own data.

Gartner's estimate that poor data quality costs organisations at least **\$12.9 million a year on average** is drawn from a 2020 research note and covers all organisations rather than retail specifically.⁵ It is cited here as directional context, not as a retail benchmark.

4. WHY PAPER AND SPREADSHEETS PERSIST

The inFlow *State of Inventory Management 2026* survey ran in March 2026 across 400 full-time warehouse, inventory, supply chain and operations professionals in 33 sectors.⁴

PRIMARY INVENTORY TOOL	SHARE OF RESPONDENTS
Spreadsheets as sole or primary tool, no dedicated software	74.2%
Spreadsheets only	52.5%
Spreadsheets combined with paper records	21.8%
Dedicated inventory management software	22%
ERP system	19%
Warehouse management system	14.5%
OPERATIONAL OUTCOME	SHARE OF RESPONDENTS
Experience stockouts at least monthly	44%
Order accuracy of 99–100%	16.2%
Order accuracy of 95–98%	54.5%
Order accuracy of 90–94%	23.8%

The sample skews towards construction (25.5% of respondents) and towards larger firms (35% employ 500 or more people), so it should not be read as a pure retail SME picture. Its value is in establishing that manual and semi-

manual stock control is not a fringe practice confined to very small operators. It is the majority condition even among organisations large enough to afford an alternative.

5. WHAT A RELATIONAL MODEL ACTUALLY FIXES

It is worth being precise about the mechanism, because the vendor framing (“software makes inventory accurate”) is not what the evidence supports. A paper or spreadsheet stock system across five branches fails in four specific ways, and a normalised relational schema with enforced constraints removes three of them outright.

5.1 The four failure modes

FAILURE MODE	WHAT IT LOOKS LIKE IN A PAPER OR SPREADSHEET ESTATE	REMOVED BY A RELATIONAL MODEL?
Referential drift	Branch 3 records stock against a product code that no longer exists in the master list; an order references a supplier that was deleted last quarter. Nothing prevents it, so nothing catches it.	Yes. A foreign key constraint makes the write fail at the moment it is attempted.
Duplicate and divergent masters	Each branch maintains its own product sheet. The same item exists as three codes with three descriptions and three cost prices. Group-level stock cannot be summed.	Yes. A single Product entity with a primary key makes divergent local copies structurally impossible.
Update anomalies	A supplier changes address. It is corrected in two branch files and missed in three. Deliveries continue to route to the old address.	Yes. Normalisation means the address is stored once and read everywhere.
Physical-to-record divergence	Stock is stolen, damaged, miscounted at goods-in, or sold without being scanned. The database faithfully records a number that is wrong.	No. This is a process and physical-control problem. No schema fixes it.

THE HONEST LIMIT OF THE CLAIM

The ECR study's 59.54% discrepancy rate was measured at retailers that already run electronic systems. Their records were still wrong more than half the time. A relational database is a necessary condition for accuracy at multi-site scale, not a sufficient one. What it delivers is the ability to *detect and attribute* the fourth failure mode, because for the first time there is a trustworthy expected value to compare the physical count against.

5.2 Mapping to the AC51049 schema

The eight entities implemented in the AC51049 project map onto the failure modes above as follows.

ENTITY	ROLE IN INTEGRITY
Supplier	Single master record; removes divergent supplier detail across branches and anchors lead-time and cost analysis.
Branch	Makes location a first-class dimension rather than a filename, which is what allows stock to be summed and compared across the estate.
Employee	Attribution. Every transaction has an owner, which is the precondition for diagnosing where discrepancies originate.
Customer	Links demand to the branch that served it, enabling branch-level demand analysis rather than estate-level averages.
Product	The single source of truth for item identity. This is the entity that kills the duplicate-masters failure mode.
Inventory	The Product-by-Branch junction. Holds the expected quantity that a physical count is measured against.
Order / OrderDetails	Header-and-line separation, which is what allows one order to hold many lines without repeating header data, and what makes line-level margin and supplier performance analysis possible.

6. AN ILLUSTRATIVE MODEL, CLEARLY LABELLED AS ILLUSTRATIVE

THIS SECTION IS AN EXTRAPOLATION, NOT A MEASUREMENT

The figures below apply published benchmarks to a hypothetical five-branch retailer. They are not outcomes of the AC51049 project and must not be presented as such. They are included to show the order of magnitude at stake, and to make the assumptions visible so they can be challenged.

The derivation is deliberately simple so that every step can be contested.

1. ECR found that 59.54% of SKUs carried a discrepancy.¹
2. Correcting records produced an average sales lift of 5.98% on the affected items.¹
3. If the lift is applied only to the revenue share of affected SKUs, the store-level revenue effect is approximately $0.5954 \times 5.98\% = 3.56\%$.

ANNUAL REVENUE ACROSS FIVE BRANCHES	REVENUE SHARE AFFECTED (59.54%)	IMPLIED ANNUAL UPLIFT AT 3.56%
£1,000,000	£595,400	£35,600
£2,500,000	£1,488,500	£89,000
£5,000,000	£2,977,000	£178,000
£10,000,000	£5,954,000	£356,000

Four assumptions in that table are doing a great deal of work, and an interviewer should be expected to attack all four:

- **The uplift decays.** ECR's second experiment showed accuracy deteriorating over a 16-week window.¹ Treating a one-off lift as a permanent annual gain overstates the benefit.
- **Discrepant SKUs are assumed to carry a proportional share of revenue.** They almost certainly do not; discrepancy concentrates in fast-moving items, which would push the figure up, while slow-moving tail items

would push it down.

- **The lift was measured after a stocktake, not after a database migration.** The two interventions are related but not the same, and the study does not license attributing the full lift to a schema change.
- **Shabani et al. show the benefit is not uniform.**² A near-efficient operator would see materially less than this. A weak one would see more.

The defensible version of the claim is therefore narrower than the table suggests: **for a five-branch retailer currently running paper stock control, the addressable prize is in the low single-digit percentages of revenue, it is concentrated in fast-moving lines, and capturing it requires an ongoing accuracy process rather than a one-off system implementation.**

7. LIMITATIONS OF THIS REVIEW

- **Source mix.** Of the five primary sources, one is peer-reviewed (Shabani et al., *IJPE*), one is a large industry-funded field study with published methodology (ECR), one is analyst research from a firm with vendor relationships (IHL), one is a vendor-run survey (inFlow), and one is a 2020 consultancy estimate (Gartner). They are not of equal evidential weight and are not treated as such above.
- **Sample bias.** The inFlow survey over-represents construction and larger employers. The ECR study covers European grocery, general merchandise and fashion, not the full retail spectrum.
- **Currency.** The Gartner figure dates from 2020 and is not inflation-adjusted here.
- **No causal claim.** Nothing in this review establishes that relational database adoption causes a measurable sales uplift. The evidence establishes that record inaccuracy is widespread and costly, and that structural data problems are one identifiable contributor.

8. SOURCES

1. Rekik, Y., Syntetos, A. and Glock, C. (2020). *Measuring the Sales Impact of Improving Inventory Records*. ECR Retail Loss, published 6 July 2020. Approximately 1m SKUs examined, 233,000 in final analysis, 7 European retailers, 100 stores, 3-year programme.
<https://www.ecrloss.com/research/grow-sales-by-improving-inventory-records/>
2. Shabani, A., Maroti, G., de Leeuw, S. and Dullaert, W. (2021). 'Inventory record inaccuracy and store-level performance', *International Journal of Production Economics*, 235, article 108111. 81 stores, 5.25m+ inventory items, network DEA model.
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3. IHL Group (2025). *Retail Inventory Crisis Persists Despite \$172 Billion in Improvements*.
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4. inFlow Inventory (2026). *State of Inventory Management 2026*. Survey of 400 full-time inventory, warehouse, supply chain and operations professionals, fieldwork March 2026, 33 sectors, 38 questions.
<https://www.inflowinventory.com/blog/state-of-inventory-management-2026/>
5. Gartner. *Data Quality: Why It Matters and How to Achieve It*, citing 2020 Gartner research on the cost of poor data quality.
<https://www.gartner.com/en/data-analytics/topics/data-quality>

Prepared as a companion note to the Relational Database Design (AC51049) project, University of Dundee. All figures are third-party industry benchmarks and are attributed above. No figure in this note is a measured result of that project.